

Replication Report: *Kondo Destruction and Multipolar Order* arXiv:1807.09258 (Lai, Nica, Hu, Gong, Paschen, Si, 2018)

Automated replication (8-artifact standard)

2026-07-19

1 Paper summary

Lai *et al.* study a spin-1 Kondo lattice model on the square lattice with *both* spin (dipole) and quadrupole local degrees of freedom,

$$H_{\text{KL}} = H_S + H_c + H_K, \quad (1)$$

$$H_S = \sum_{ij} \left[J_{ij} (\mathbf{S}_i \cdot \mathbf{S}_j) + K_{ij} (\mathbf{S}_i \cdot \mathbf{S}_j)^2 \right], \quad (2)$$

$$H_K = \sum_j (J_K^I \mathbf{S}_j \cdot \mathbf{s}_{c,j} + J_K^{II} \mathbf{Q}_j \cdot \mathbf{q}_{c,j}). \quad (3)$$

Using DMRG they establish a (π, π) *antiferroquadrupolar* (AFQ) phase for $J_1 = 1$, $K_1 = 1.2$, $J_2 = 0$, $K_2 < 0$, characterized by a staggered $\langle Q_j^{x^2-y^2} \rangle \sim (-1)^j$. They then derive a quantum non-linear sigma model (NL σ M) for the AFQ order using an SU(3) time-reversal-invariant director (d -vector) representation and a combined boson-fermion renormalization group (RG) analysis, concluding that the Kondo couplings are *exactly marginal* in the AFQ phase (Kondo destruction): the Kondo vertex carries positive powers of $\sqrt{\Lambda/K_F}$ and is asymptotically suppressed.

The “multipolar texture” here is the spatial pattern of complex directors $\mathbf{d}_j = (d_j^x, d_j^y, d_j^z)$ encoding the quadrupolar order: $\mathbf{d}_A^{\text{gs}} = (1, 0, 0)^T$, $\mathbf{d}_B^{\text{gs}} = (0, 1, 0)^T$ on the two sublattices (Fig. 2a of the paper).

2 Scope of this replication

Full DMRG (4000 SU(2) states, $L_y = 8$ cylinders) and the field-theoretic RG are out of scope for a laptop. We replicate the *exactly checkable algebraic core* plus a *tractable finite-cluster ED surrogate* of the DMRG structure-factor claim:

- the operator algebra (biquadratic identities, quadrupole operators),
- the d -vector $\rightarrow$ spin/quadrupole maps (Eqs. S16–S22),
- the ground-state director pattern and its energetics (Eq. S23/S24),
- the SU(3) Gell-Mann rotation structure and the flat (marginal) direction,
- small-cluster ED of the spin-1 BLBQ model showing the (π, π) AFQ peak.

3 Checked claims and results

C1 — Biquadratic operator identities (exact)

The paper’s algebraic identities

$$(\mathbf{S}_i \cdot \mathbf{S}_j)^2 = \frac{1}{2} \mathbf{Q}_i \cdot \mathbf{Q}_j - \frac{1}{2} \mathbf{S}_i \cdot \mathbf{S}_j + \frac{1}{3} \mathbf{S}_i^2 \mathbf{S}_j^2, \quad \mathbf{Q}_i \cdot \mathbf{Q}_j = 2(\mathbf{S}_i \cdot \mathbf{S}_j)^2 + \mathbf{S}_i \cdot \mathbf{S}_j - \frac{8}{3}$$

were verified operator-by-operator on the 9-dimensional two-site spin-1 space: residuals $\sim 10^{-16}$ – 10^{-15} . **PASS**.

C2 — d -vector maps (Eqs. S16–S22)

For 200 random normalized directors we compared $\langle S^a \rangle, \langle Q^a \rangle$ from the 3×3 operators (in the TRI basis, Eq. S10) against the closed-form d -vector formulas $S^\alpha = -i\epsilon^{\alpha\beta\gamma} \bar{d}^\beta d^\gamma$, $Q^{x^2-y^2} = -|d^x|^2 + |d^y|^2$, etc. Max discrepancy 5.6×10^{-16} . The ground-state directors carry *zero* dipole moment. **PASS**.

C3 — (π, π) -AFQ ground state

$\mathbf{d}_A = (1, 0, 0)$, $\mathbf{d}_B = (0, 1, 0)$ give $\langle Q_A^{x^2-y^2} \rangle = -1$, $\langle Q_B^{x^2-y^2} \rangle = +1$ (staggered $(-1)^j$). At the SU(3) point the nearest-neighbor bond energy $\propto |\mathbf{d}_i \cdot \bar{\mathbf{d}}_j|^2$ is minimized ($= 0$ for orthogonal directors) while the second-neighbor (same-sublattice, parallel) overlap is maximized ($= 1$), matching the paper’s stated minimization/maximization requirement. **PASS**.

C4 — Gell-Mann rotations and marginality structure

The four generators (Eq. S25) are Hermitian $\Rightarrow U(\phi) = e^{i\sum \lambda_p \phi_p}$ is unitary (residual 9×10^{-16}). For 200 random rotations at the SU(3) point, the two-director energy is invariant to $\sim 10^{-15}$: the SU(3) rotations are exact flat directions (Goldstone modes), the geometric substrate for the “exactly marginal Kondo coupling” claim. **PASS** (structural; the full RG scaling $S_K/S_c \propto \sqrt{\Lambda/K_F}$ is analytic and reproduced in the report text, not independently re-derived numerically).

C5 — SU(3) multiplet structure of the two-site spectrum

At $J = K$ the two-site Hamiltonian has exactly two distinct eigenvalues with degeneracies $\{6, 3\}$ — the $\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}$ decomposition of SU(3). Away from $J = K$ ($J=1, K=1.2$) the levels split (three distinct values), confirming the lowering SU(3) $\rightarrow$ SU(2). **PASS**.

C6 — Finite-cluster ED of the (π, π) AFQ (DMRG surrogate)

Exact diagonalization of the spin-1 BLBQ model at $J_1=1, K_1=1.2, J_2=0, K_2=-0.3$ on periodic 2×2 and 2×4 clusters (sparse Lanczos, cross-checked by dense **eigh**): the quadrupolar structure factor $m_Q^2(\mathbf{q})$ peaks at $\mathbf{q} = (\pi, \pi)$ and dominates the dipolar m_S^2 , whose weak peaks sit at $(\pi, 0)/(0, \pi)$ — exactly the pattern of Figs. 1 and S1.

PASS (qualitative; magnitudes out of scope for a laptop cluster).

cluster	$m_Q^2(\pi, \pi)$	$m_S^2(\pi, \pi)$	$m_S^2(\pi, 0)$	quad peak	spin peak
2×2	1.976	0.384	0.808	(π, π)	$(\pi, 0)$
2×4	1.256	0.218	0.487	(π, π)	$(\pi, 0)$

Table 1: ED structure factors; dominant m_Q^2 at (π, π) , weak m_S^2 at $(\pi, 0)/(0, \pi)$, matching the paper qualitatively. Magnitudes are finite-size and are *not* claimed to match the thermodynamic-limit DMRG order parameters.

4 Verdict

Verdict: REPLICATED (core algebra + qualitative numerics).

Coverage: 7/10 — all analytic/algebraic claims and the qualitative DMRG structure-factor result reproduced; full DMRG magnitudes and the field-theoretic RG derivation not independently reproduced (out of scope).

Agreement: 9/10 — every in-scope quantitative check matches to machine precision or with the correct qualitative peak structure.

All 12 algebraic sub-checks pass to $\lesssim 10^{-15}$; the ED surrogate reproduces the (π, π) -AFQ dominance on both clusters. No contradictions with the paper were found within scope.